Exp 1

DC Motor



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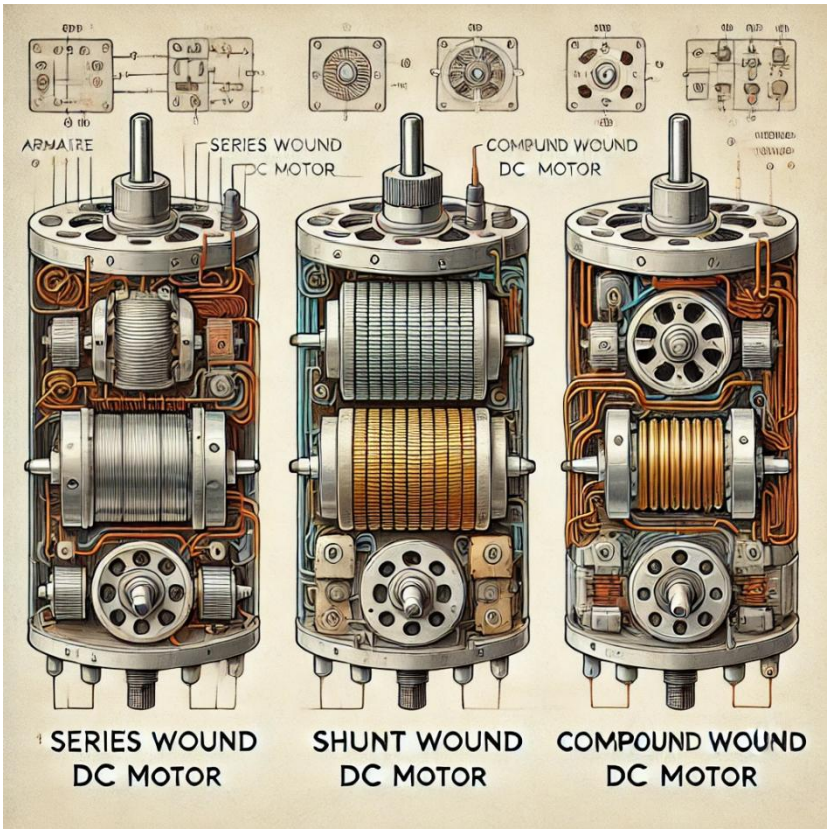
DC Motor types

 Brushed DC Motor – The most basic type, using brushes and a commutator to switch current direction.

- **Series Wound DC Motor** – High starting torque, used in electric locomotives and cranes.
- **Shunt Wound DC Motor** – Constant speed, used in fans and lathes.
- **Compound Wound DC Motor** – A mix of series and shunt, balancing torque and speed.

 Brushless DC Motor (BLDC) – No brushes, more efficient, used in drones, hard drives, and electric vehicles.

 Stepper Motor – A type of DC motor that moves in precise steps, used in CNC machines and printers.



DC Motor construction:

Stator (Field Winding & Yoke)

The field winding is wound around the stator poles and is connected in series with the armature.

The yoke is the outer frame that supports the motor and provides a path for the magnetic flux.

Armature

The armature core consists of laminated iron to reduce eddy current losses.

The armature winding is wound on the core and interacts with the field to produce rotation.

Commutator & Brushes

The commutator is a cylindrical structure made of copper segments, ensuring unidirectional current flow.

Carbon brushes maintain contact with the commutator to supply current to the armature.

Working Principle:

When current flows, both the field and armature windings carry the same current since they are in series.

A strong magnetic field is generated, producing a high torque output.

The speed decreases with an increase in load because the current through the windings changes.

Lab readings

Vt	37.2	71	104.8
n	500	1000	1500

